

Literature Review

CyberSense AI: AI-Powered Human Cyber Behaviour Prediction System

1. Introduction

Most cybersecurity tools detect attacks only after they occur, yet a large share of breaches trace back to human decisions — password reuse, clicking malicious links, replying to fraudulent SMS, or connecting unknown devices. CyberSense AI addresses this by using AI/ML to *predict* elevated human cyber-risk before an incident occurs, across multiple behavior categories (credential, phishing, social engineering, network, and device risk), and by explaining every prediction using Explainable AI (XAI) so users can act on it.

2. Methodology & Research Background

Papers were sourced from Elsevier, Springer, and IEEE (2023–2026), covering behavioural cybersecurity, insider threat detection, phishing detection, explainable AI, authentication behavior, and behavioral drift detection. All papers were read in full and verified against source before inclusion. Core concepts underpinning this review: User Behavior Analytics (UBA) baselining normal behavior to flag deviations; Explainable AI (XAI) — SHAP/LIME-based attribution of predictions to features; Insider Threat Detection — modeling employee behavior over time, historically benchmarked on the CERT dataset; Phishing Detection across email, SMS, and URL channels; Behavioural Drift Detection tracking gradual change in a user's habits over time.

3. Reviewed Literature

#	Category	Paper (Year, Publisher)	Method	Dataset	Key Result / Finding
1	Human-Centered	Predicting cyber-criminal characteristics (Elsevier)	Classical ML on behavioral/demographic data	Survey data	Offender profiling; not end-user risk
2	Human-Centered	Human-factor security in	Behavioral simulation review	Conceptual	Discusses federated

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		cyberfinance (Springer, 2026)			learning, blockchain
3	Human-Centered	Human factors in cybersecurity: interdisciplinary review (Springer)	Literature synthesis	N/A	Framework proposal, no predictive ML
4	Human-Centered	Leveraging human factors: integrated methodology (Springer)	Mixed-methods	N/A	Conceptual grounding only
5	Insider Threat	BTITD (2026, Springer)	Dual-stream TimesNet+BiLSTM (separates behavior/timestamp)	CERT r4.2	F1=97.56% binary, 94.66% multi-class
6	Insider Threat	FusionITD (2026, Springer)	GNN + semantic (RAG) fusion of behavior + email/IM content	CERT r4.2/r5.2	+5% AUC; semantic fusion mainly cuts false positives
7	Insider Threat	ATHITD (2026, Elsevier <i>Computers & Security</i>)	Attention-based temporal heterogeneous GNN	CERT r4.2, LANL	F1=0.96–0.97
8	Insider Threat	BRITD (2026, Springer <i>Cybersecurity</i>)	Stacked BiLSTM, time-aware, user-adaptive	CMU CERT	AUC=0.9730
9	Insider Threat	Optimising Insider Threat	BiLSTM vs. RNN/LSTM comparison	CERT r4.2	BiLSTM outperforms

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		Prediction (2024, Springer)			unidirectional models
10	Insider Threat	UEBA hybrid (Springer)	Attention Bi-LSTM + ResNet	CERT r4.2	F1 up to 0.9752
11	Insider Threat	Supervised ML insider detection (2023, Springer)	Supervised ML vs. LSTM Autoencoder	CERT r4.2	LSTM Autoencoder strongest baseline
12	Insider Threat	Forensic techniques for insider threats (2026, Springer)	Time-based activity + hybrid anomaly detection	CERT Email dataset	Strong for post-incident investigation
13	Insider Threat	SPEDIA dataset paper (2025, Elsevier)	Introduces SPEDIA — Wazuh SIEM-agent logs, harmonized with CERT schema	SPEDIA (self)	25 curated attributes; fully reproducible pipeline — this project's actual dataset
14	Insider Threat	Session-Based Detection using CERT & SPEDIA (2026, SSRN)	36-feature session vectors; supervised + unsupervised ML compared	CERT r4.2, validated on SPEDIA	Bridges CERT-based literature to SPEDIA directly
15	Explainable AI	Rule Extraction & Interaction-Aware XAI for Malware (Springer)	Rule extraction from AI malware classifiers	N/A	Malware-specific, not behavior prediction
16	Explainable AI	Explainable Cyber Threat Intelligence	XAI-ranked threat prioritization	N/A	Framework-level, limited benchmarking

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		Framework (Springer)			
17	Explainable AI	Dynamic XAI risk management for cyber insurability (Springer)	SHAP-based dynamic org. risk scoring	Organizational data	Adaptive + explainable, but organization-level
18	Explainable AI	PRISMA XAI systematic review (565→67 papers)	Meta-analysis	N/A	SHAP/LIME dominant; explanations can leak sensitive features
19	SMS/Smishing	Smishing detection via BERT (2025, Springer)	BERT + NLP	SMS smishing corpus	Strong smishing (phishing subtype) detection
20	SMS/Smishing	Cost-Sensitive SMS Spam Detection (2026, Springer)	3-step preprocessing + SVM	SMS Spam Collection	Explicit class-imbalance handling
21	SMS/Smishing	ALBERT-BiLSTM cross-attention (2025, Elsevier)	ALBERT-BiLSTM + knowledge distillation	SMS Spam Collection	Strong cross-domain generalization
22	Email Phishing	Explainable RoBERTa phishing detector (2026, Elsevier)	RoBERTa + LIME	Phishing Emails (Kaggle)	98.45% accuracy, explainable

#	Category	Paper (Year, Publisher)	Method	Dataset	Key Result / Finding
24	Malicious URL	DomURLs_BERT (2026, Springer)	Pre-trained BERT for domains/URLs	6 benchmark sets (incl. this project's URL dataset)	98.5–100% accuracy
25	Authentication	PassREfinder-FL (2025, Elsevier)	GNN link prediction + federated learning over password-reuse graph	360M breached accounts	F1=0.9153; privacy-preserving
26	Authentication	Genetic Programming password security (Springer)	GP symbolic classifier	Password Strength Classifier (Kaggle)	0.99 accuracy
27	Behavioral Drift	Frouros (Elsevier)	Open-source drift-detection library	General ML streams	Not cybersecurity-specific
28	Behavioral Drift	Concept drift in behavioral biometrics (Elsevier)	Drift detection for biometric auth	Biometric data	Domain-specific, not general cyber-risk

4. Research Gap

- **Single-risk focus** — every system above targets one risk type only; none unify multiple categories into one profile.
- **Reactive, not predictive** — most insider-threat systems classify after the fact rather than forecasting rising risk.
- **XAI applied narrowly** — SHAP/LIME used for single-task classifiers, not continuous multi-category risk scores.
- **Drift detection rarely applied to cyber-hygiene** — mature for biometrics/data streams, not personal security habits.

- **Personalized coaching is essentially absent** — systems stop at a flag or static warning.
- **Structural and content-based signals are never combined** — e.g., PassREfinder-FL's graph approach vs. BERT-based content classifiers.

5. CyberSense AI's Contribution

CyberSense AI integrates SMS, email, URL, credential, and device/network risk signals into one continuously updated, LSTM-based predictive profile per user, paired with SHAP-based explainability across all categories and behavioral drift detection over time — directly addressing every gap above. Where the literature offers strong individual components (LSTM/BiLSTM sequence modeling, BERT-based phishing/SMS/URL classifiers, SHAP explainability, graph-based credential risk), CyberSense AI is the first to combine them into one adaptive, explainable, coaching-oriented system, built on SPEDIA rather than the more limited CERT dataset.

6. Ethical Considerations & Future Directions

Ethics: informed consent for behavioral monitoring; data minimization (following PassREfinder-FL's federated approach); explanations should reveal risk *categories*, not raw identifying details (per PRISMA XAI review); and a clear ethical line between coaching users and using predictions punitively.

Future work: a shared transformer backbone across all risk categories; federated learning for cross-organizational deployment; real-time streaming inference; LLM-based personalized coaching; and evaluation on real (not only synthetic) organizational data.

7. Conclusion

This review confirms that CyberSense AI's core techniques — LSTM/BiLSTM sequence modeling, BERT/RoBERTa-based phishing detection, SHAP-based explainability, and graph-based credential risk — are each individually well-validated in recent (2023–2026) Elsevier and Springer literature. No reviewed system, however, integrates them into a single explainable, multi-category, adaptive-coaching platform — the specific gap CyberSense AI is designed to fill.